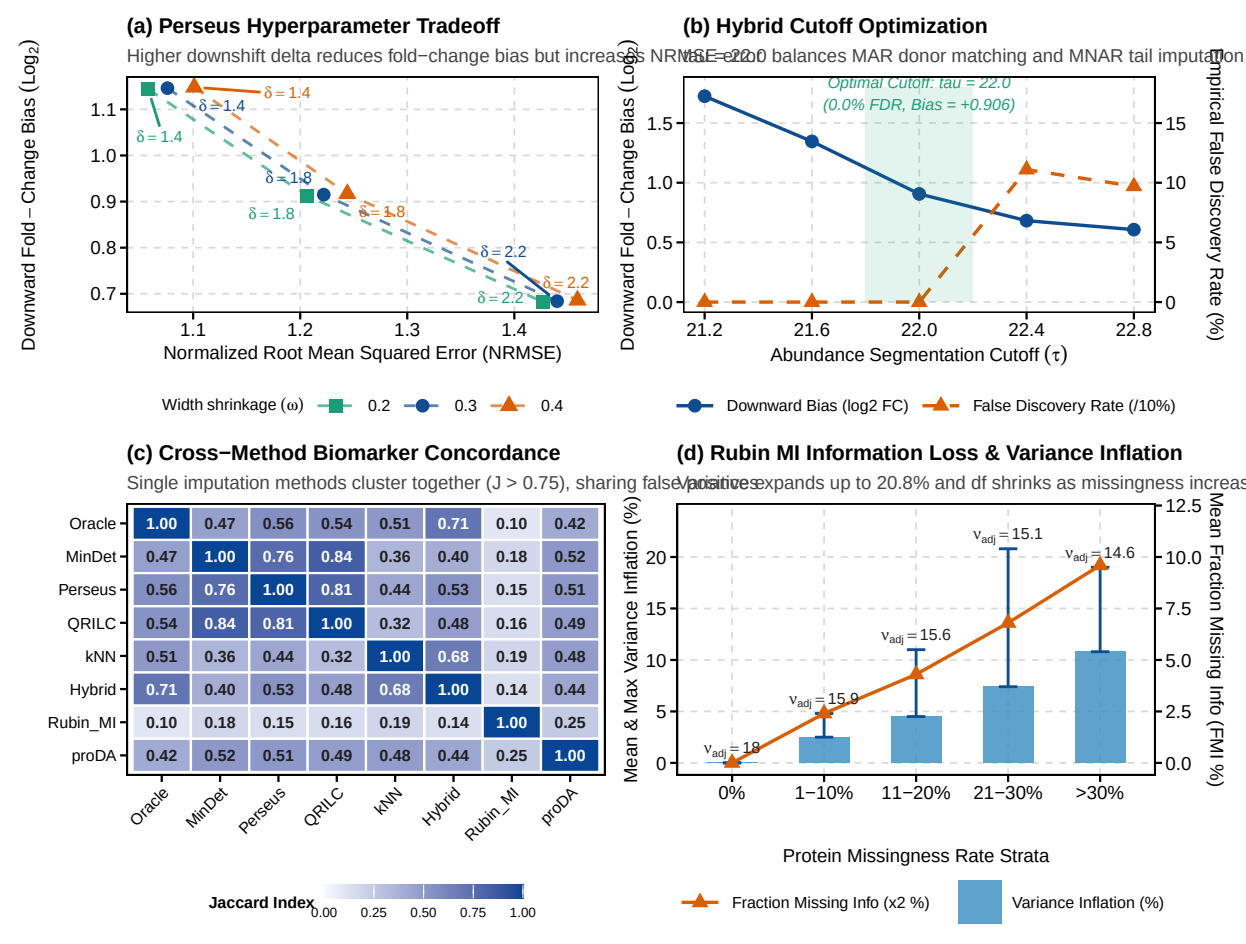
(a) Perseus Hyperparameter Tradeoff
Higher downshift delta reduces fold−change bias but increases NRMSE
Downward Fold − Change Bias (Log2)
Normalized Root Mean Squared Error (NRMSE)
δ = 1.4
δ = 1.8
δ = 2.2
Width shrinkage (ω) 0.2 0.3 0.4
(b) Hybrid Cutoff Optimization
MAE = 22.0 balances MAR donor matching and MNAR tail imputation
Optimal Cutoff: tau = 22.0
(0.0% FDR, Bias = +0.906)
Downward Fold − Change Bias (Log2)
Empirical False Discovery Rate (%)
Abundance Segmentation Cutoff (τ)
Downward Bias (log2 FC)
False Discovery Rate (/10%)
(c) Cross−Method Biomarker Concordance
Single imputation methods cluster together (J > 0.75), sharing false positives
Oracle 1.00 0.47 0.56 0.54 0.51 0.71 0.10 0.42
MinDet 0.47 1.00 0.76 0.84 0.36 0.40 0.18 0.52
Perseus 0.56 0.76 1.00 0.81 0.44 0.53 0.15 0.51
QRILC 0.54 0.84 0.81 1.00 0.32 0.48 0.16 0.49
kNN 0.51 0.36 0.44 0.32 1.00 0.68 0.19 0.48
Hybrid 0.71 0.40 0.53 0.48 0.68 1.00 0.14 0.44
Rubin_MI 0.10 0.18 0.15 0.16 0.19 0.14 1.00 0.25
proDA 0.42 0.52 0.51 0.49 0.48 0.44 0.25 1.00
Jaccard Index
(d) Rubin MI Information Loss & Variance Inflation
Variance expands up to 20.8% and df shrinks as missingness increas
Mean & Max Variance Inflation (%)
Mean Fraction Missing Info (FMI %)
ν_adj = 18
ν_adj = 15.9
ν_adj = 15.6
ν_adj = 15.1
ν_adj = 14.6
0% 1−10% 11−20% 21−30% >30%
Protein Missingness Rate Strata
Fraction Missing Info (x2 %)
Variance Inflation (%)